Contact

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Top Skills

Technical Site Leadership & Capital Strategy

Federal R&D Leadership (PI & Program Management)

Hardware Scale-Up & Prototype-to-Production

Languages

English (Native or Bilingual)

Spanish (Professional Working)

Anthony Palumbo

Technical Site Lead (PhD) | PI & PM ($4.6M SBIR) | High-Precision Optics & Advanced Ceramics | R&D Scale-Up & Facility Strategy

Greater Boston

Summary

I lead the development and scale-up of high precision optical and advanced ceramic components, translating complex R&D into deployable, mission critical hardware for defense and aerospace systems.My focus is building the technical and operational frameworks required to move advanced materials from research into reliable, production ready platforms, reducing risk across the full hardware lifecycle TECHNICAL & PROGRAM LEADERSHIPI currently serve as Technical Site Lead, directing the mirror fabrication vertical within a $30M+ multi phase portfolio. I operate at the intersection of technical execution and operational strategy for government sponsored programs.As Principal Investigator and Program Manager, I manage $4.6M in SBIR/STTR funding across multiple high stakes initiatives, including:• Phase I to Phase II prime advancement ($1.4M+ total)• Phase II subcontract leadership ($3.2M combined)• Transitioning R&D into production relevant systems aligned with national security objectives.SELECTED IMPACT:• Delivered prototype mirrors and established scalable manufacturing workflows for high reliability aerospace hardware• Led the mirror prototype vertical across six major programs ($30M+), bridging R&D, engineering, and production• Directed the design and commissioning of an 11,000 sq. ft. advanced optics facility, including 8 major production and metrology systems• Conceived and secured Phase I SBIR funding and advanced to Phase II through independent technical leadership.CORE CAPABILITIES:Technical Site Leadership | Hardware R&D Scale-Up | Program & Portfolio Management | Advanced Manufacturing | Facility & Capital Strategy | Mission Critical Defense Systems | SBIR/STTR Lifecycle

Experience

Advanced Silicon Carbide Materials

4 years 8 months

Principal Investigator & Technical Site Lead (Director of Technical Operations)
January 2025 - Present (1 year 4 months)

- Facility Architecture: Spearheaded the design and build-out of an 11,000 sq. ft. advanced optics facility; deployed $4M+ in capital assets to establish pilot-scale fabrication.

- Federal Program PI: PI for $4.6M in SBIR/STTR funding; originated and led a Phase I program through competitive down-selection into Phase II ($1.2M).

- Portfolio Leadership: Directing specialized mirror fabrication workstreams within a $30M multi-project R&D portfolio.

- Operational Scaling: Leading multidisciplinary engineering teams to transition advanced ceramics and optics from lab-scale to pilot production.

- Strategic Roadmapping: Defining technology and commercialization pathways, including successful progression of federally funded programs from early-stage R&D through Phase II prototype development.

Principal Investigator (Technical Program Manager)
January 2024 - December 2024 (1 year)

- Venture Funding & Strategy: Secured $240k in Phase I SBIR funding as Principal Investigator; managed the full lifecycle from proposal development to technical execution and prototype delivery.

- Technical Architecture: Established and matured initial optics fabrication and testing capabilities, developing the process documentation and workflows required for Phase II scaling.

- Program Management: Served as the primary technical and programmatic liaison for government stakeholders, translating federal program objectives into executable R&D milestones.

- Cross-Functional Leadership: Directed multidisciplinary teams to deliver high-precision prototypes on aggressive schedules, ensuring alignment with future commercialization and Phase II transition pathways.

Senior Research Scientist (R&D Materials Specialist)
September 2021 - December 2023 (2 years 4 months)

Lihue, HI

- CVD Manufacturing & Process R&D: Led materials research and process development for CVD-based optics and ceramics; translated fundamental feasibility studies into applied manufacturing workflows for multiple federal programs.

- Data Science & Metrology: Engineered internal materials databases and implemented Statistical Process Control (SPC) and analytical modeling to drive R&D decision-making and yield optimization.

- Business Development & Capture: Spearheaded SBIR solicitation analysis and technical proposal writing, directly supporting the acquisition of new government contracts and portfolio growth.

- Technical Stakeholder Engagement: Represented the organization as a Subject Matter Expert (SME) at Technical Interchange Meetings (TIMs) and tier-1 industry conferences, communicating complex materials capabilities to federal customers.

- Systems Engineering Support: Executed government contract requirements through rigorous component selection, performance analysis, and cross-functional technical alignment.

Scientific Design Company, Inc.
Advanced Materials R&D Engineer (Materials Characterization Specialist)
May 2019 - November 2021 (2 years 7 months)
Little Ferry, NJ

- Synthesis & Industrial Scale-Up: Engineered advanced alumina carrier materials for industrial-scale catalysis; translated patent analysis and literature reviews into high-performance synthesis recipes.

- Automated Image Analysis & Computer Vision: Pioneered a self-directed project utilizing ImageJ-based computer vision to quantify sub-micron particle geometries and size distributions; correlated silver microstructure with catalyst performance to drive R&D optimization.

- High-Resolution Characterization: Served as the site expert in advanced materials analysis, utilizing HR-TEM, SEM, and Electron Diffraction to analyze crystalline structures and atomic-scale features.

- Data-Driven R&D: Applied mechanical testing and absorption measurements to guide high-stakes manufacturing decisions; translated complex experimental data into actionable insights for executive leadership.

- Technical Mentorship: Developed and led training programs for R&D staff on characterization techniques and analytical workflows, establishing site-wide best practices for lab safety and quality.

Stevens Institute of Technology
Graduate Research Lead (NASA & NSF Fellow)
May 2013 - August 2021 (8 years 4 months)
Hoboken, NJ

- Technology Commercialization: Served as Entrepreneurial Lead and Co-PI for a $50,000 NSF I-Corps grant; led the market-fit analysis and technical transition of CVD nanomaterials from laboratory research toward startup-ready energy and sensing applications.

- 8-Year Technical Trajectory: Engineered first-of-kind CVD nanomaterials for flexible electronics and biosensing; managed the full lifecycle from synthesis to device integration and characterization across 8 years of advancing R&D.

- Prestigious Fellowships: Recognized for technical excellence and program leadership with the NASA NJ Space Grant Fellowship and three University Provost Excellence Doctoral Fellowships.

- Infrastructure & Mentorship: Architected training and safety protocols for 30+ researchers and authored SOPs for multi-user Atomic Force Microscopy (AFM) systems

National Science Foundation (NSF)
Co-PI & Entrepreneurial Leader (Deep Tech Commercialization)
February 2019 - August 2020 (1 year 7 months)
New York, NY

- Venture Strategy & Capital: Secured $50,000 in NSF funding by architecting the winning proposal, budget, and technical abstract to drive customer discovery and commercialization.

- Market-Fit Validation: Led a cross-functional team through 100+ customer discovery interviews with industry executives to validate technology-market fit and develop a scalable Business Model Canvas.

- Ecosystem Building: Translated federal research into a commercially actionable plan, establishing a strategic network of industry partners and collaborators to accelerate technology adoption.

- Strategic Leadership: Applied rigorous entrepreneurial training to de-risk early-stage R&D, positioning lab-scale innovation for Phase II transition and market entry.

Education

Stevens Institute of Technology
Doctor of Philosophy, Mechanical Engineering

Stevens Institute of Technology
Master of Engineering, Mechanical Engineering

Stevens Institute of Technology
Bachelor of Engineering, Mechanical Engineering

New York University
Bachelor of Science, Physics